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Hardware Description with SystemVerilog



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Today's Goals

After this lecture, you can

- explain why an HDL describes **hardware, not software**
- describe a circuit as **register + next-state logic**
- name the main **HDL families** and their trade-offs
- translate your **two-process VHDL** into SystemVerilog
- read and modernize plain **Verilog**
- avoid **latches, races, and width errors**
- write self-checking **SystemVerilog** and **cocotb** testbenches
- **lint, simulate, and synthesize** in IIC-OSIC-TOOLS



Chapter 2

Lecture notes: Chapter 2 of iic-jku.github.io/vlsi-design

Recap: Hardware, Not Software



MCU Software vs. Hardware Description

	Software for an MCU (C)	Hardware description (HDL)
What is described	instructions for a fixed processor	the circuit itself
Execution	one instruction after another	all blocks concurrently, all the time
Time	implicit, follows from the instruction count	explicit, registers update at clock edges
A variable is	a memory location or CPU register	a wire or a flip-flop
A loop is	a repetition in time	a repetition in space (unrolled)
A function call is	a jump to shared code	a separate copy of the hardware
Cost	memory and clock cycles	area, power, maximum clock frequency
Fixing a bug	a firmware update	for an ASIC, a new mask set

Think hardware first, then write the code.

Computing in Time and in Space

C on an MCU

```
uint8_t popcount(uint8_t x)
{
  uint8_t count = 0;
  for (int i = 0; i < 8; i++)
    count += (x >> i) & 1;
  return count;
}
```

- How many **clock cycles**?
- What happens with **64 bits**?

SystemVerilog

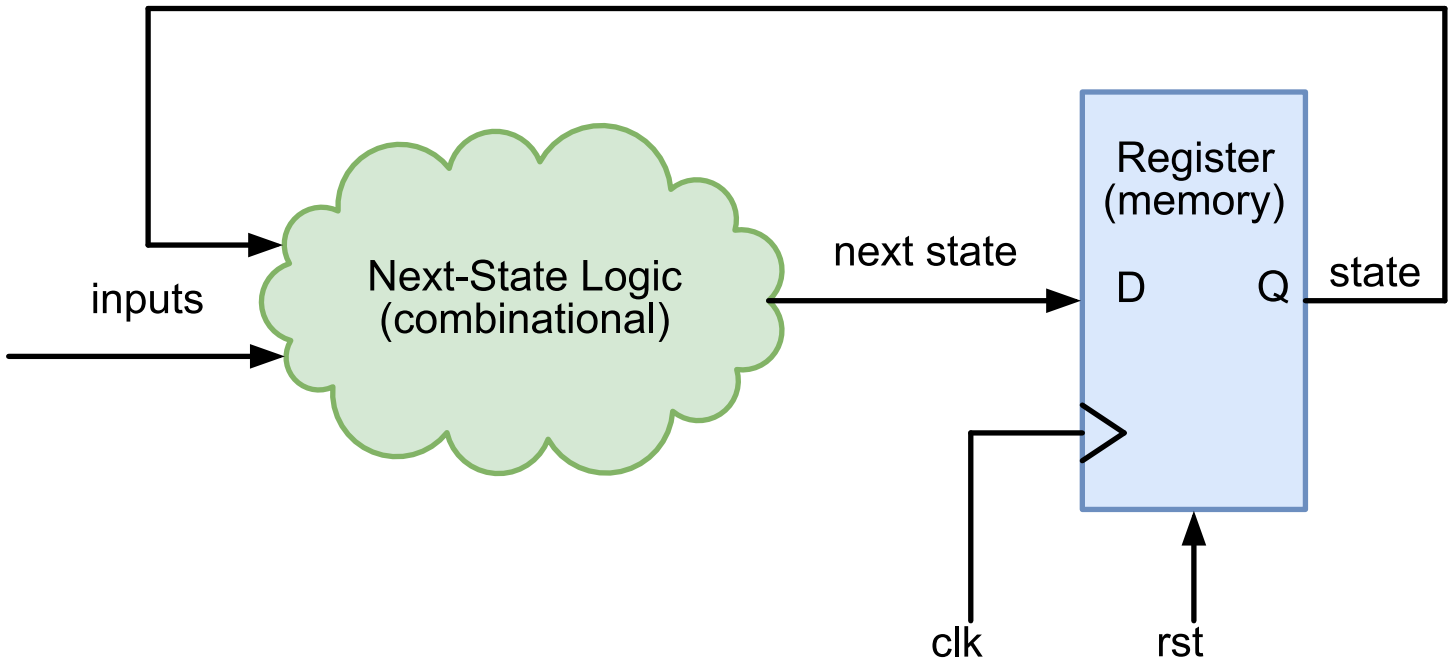
```
module popcount (
  input  logic [7:0] x_i,
  output logic [3:0] count_o
);
  always_comb begin
    count_o = '0;
    for (int i = 0; i < 8; i++) begin
      count_o = count_o + 4'(x_i[i]);
    end
  end
endmodule
```

- What does synthesis build from the **loop**?

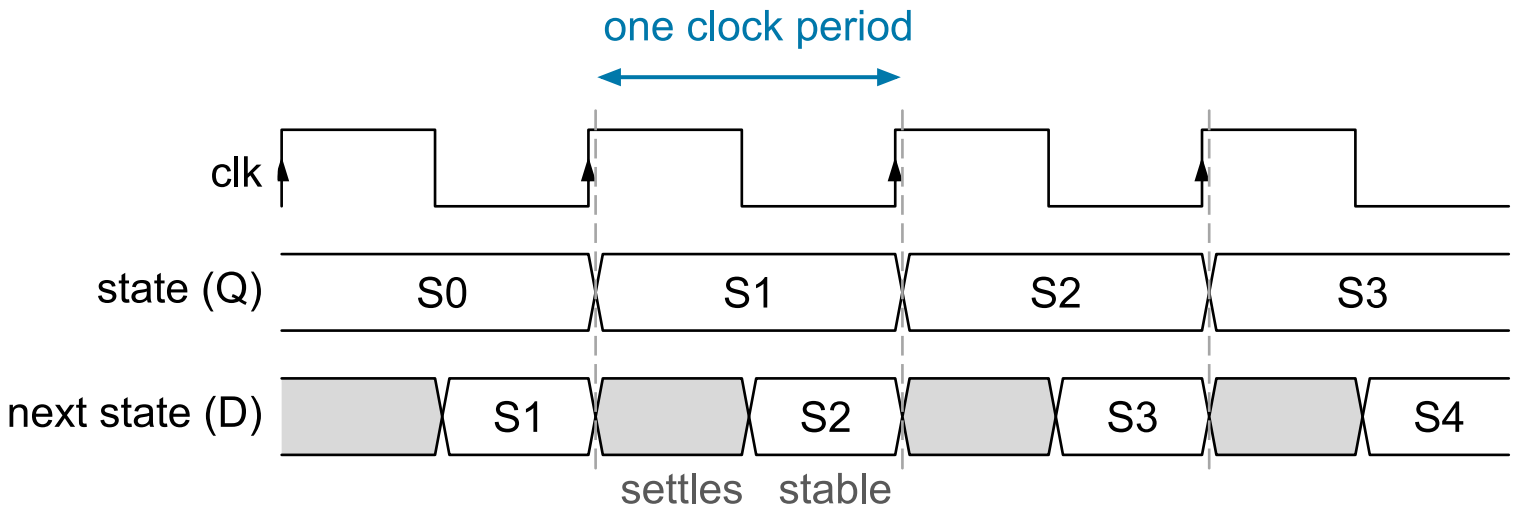
Exercise (Section 2.2.1): the three questions, then synthesize `popcount.sv`.

Synchronous Sequential Design

- **Register**: edge-triggered flip-flops store the state
- **Next-state logic**: combinational, from state and inputs
- **Rising edge**: the register loads the next state
- Outputs from the state (**Moore**) or also from the inputs (**Mealy**)



One Clock Period to Compute



- Flip-flop outputs change **only at the rising edge**
- The logic sees **stable inputs** and has **one period** to compute, minus the setup time
- Clock period > slowest path: checked by **static timing analysis** (Chapter 5)
- Asynchronous inputs need a **synchronizer** first (Chapter 9)

The Two-Process Style in VHDL

```vhdl
-- Register process: only reset and clock
reg_proc: process(rst_i, clk_i) is
begin
  if rst_i = '1' then
    state <= '0';
  elsif rising_edge(clk_i) then
    state <= next_state;
  end if;
end process reg_proc;

-- Combinational process: all signals read
next_state_logic: process(state, toggle_i) is
begin
  -- default assignment
  next_state <= state;
  -- next_state from state and inputs
  if toggle_i = '1' then
    next_state <= not state;
  end if;
end process next_state_logic;
```

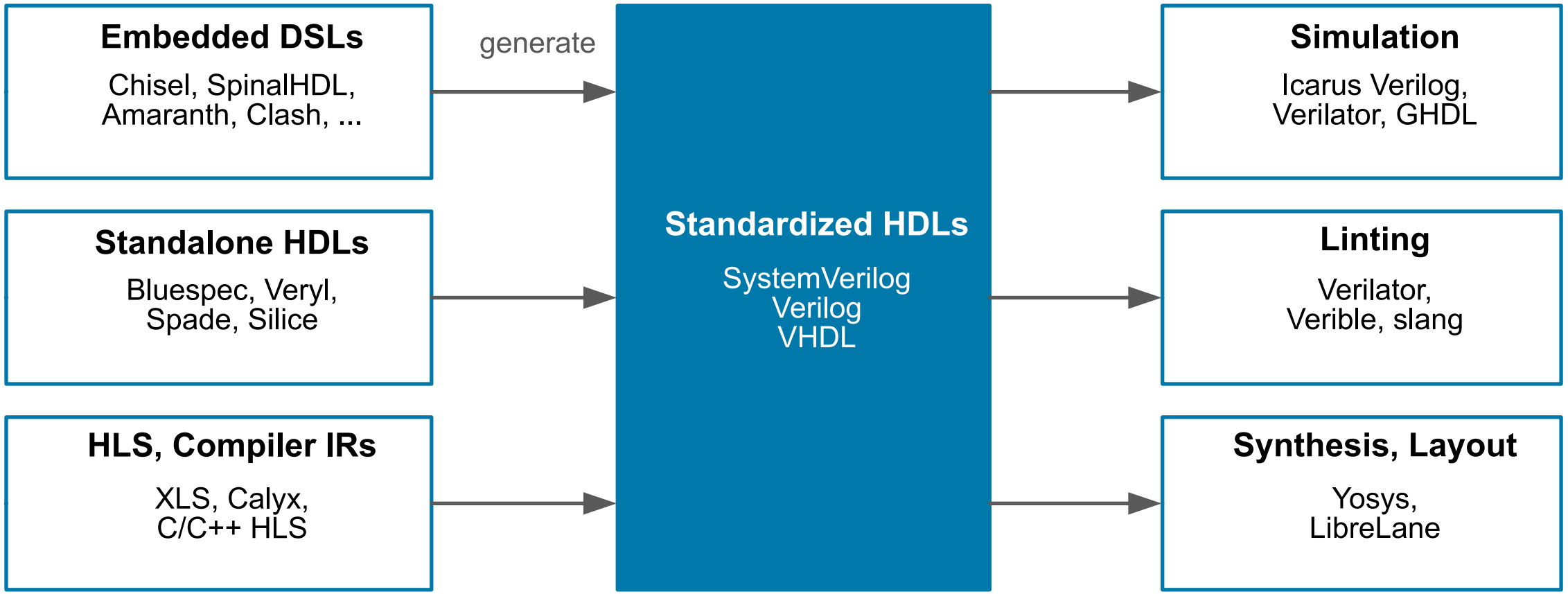
Three rules:

- **Register process**: only **reset and clock** in the sensitivity list, no logic
- **Combinational process**: **all signals it reads** in the sensitivity list (VHDL-2008: `process(all)`)
- **Default assignment first**, then all decisions: no latch

The HDL Landscape



Four Families of HDLs



All alternatives generate **(System)Verilog**, the exchange format of the industry

Comparing the Families

	Standardized HDLs	Embedded DSLs	Standalone HDLs	HLS
Adoption	very high, all EDA tools, IP, jobs	niche (Rocket Chip, VexRiscv)	research, early adopters	data paths in commercial flows
Tool support	every tool	host toolchain + generated Verilog	one compiler	mostly commercial
Productivity	RTL, limited generators	high: generators	medium to high	highest: algorithms
Error checking	VHDL strongly typed, SV needs lint	host type system	strong types (Bluespec: no races)	timing implicit
Debugging	names match	generated names	like embedded DSLs	RTL hard to relate
Long-term risk	decades of stability	small projects, some dormant	early stage	vendor lock-in

Why SystemVerilog in This Course?

- **Industry standard** (IEEE 1800-2023): every EDA tool, IP vendor, and foundry flow
- **Open-source tools**: Verilator, Icarus Verilog, Yosys, LibreLane, slang, Verible, cocotb
- **You have to read it anyway**: netlists, gate-level simulations, generated code
- **cocotb for verification**: testbenches in Python, SystemVerilog for the RTL
- **A good foundation**: Chisel or Amaranth is quick to learn later

VHDL vs. SystemVerilog

Aspect	VHDL	SystemVerilog
Standard	IEEE 1076-2019	IEEE 1800-2023 (includes Verilog)
Typing	strong, explicit conversions	weak, silent extension and truncation
Verbosity	verbose	compact
Case	case-insensitive	case-sensitive (`clk` and `Clk` differ)
Semantics	delta cycles, order never matters	races if coding rules are broken
Verification	libraries (OSVVM, UVVM)	built in: classes, SVA, coverage
Typical use	FPGA, aerospace, Europe, academia	ASIC design and verification worldwide
Open-source tools	GHDL	Verilator, Icarus Verilog, Yosys, slang

VHDL protects you with types. SystemVerilog relies on **discipline and the linter**.

Cheat-Sheets and Guidelines

- **SystemVerilog** and **Verilog cheat-sheets** in the AMS chip template
  - `doc/verilog`: syntax references, not style guides, lint what you copy
- **Circuit Designer's Etiquette**: VHDL/Verilog Coding Guide
- **lowRISC Verilog Coding Style Guide**: most of our naming
- **IEEE 1800-2023**: free through the IEEE GET Program
- Books:
  - Harris and Harris: SystemVerilog and VHDL side by side
  - Sutherland: RTL modeling with SystemVerilog
  - Sutherland and Mills: 101 gotchas

Cheat-Sheets

Etiquette

SystemVerilog for VHDL Designers



VHDL and SystemVerilog at a Glance

Concept	VHDL	SystemVerilog
Design unit	`entity` + `architecture`	`module ... endmodule`
Parameter	`WIDTH : positive := 8`	`parameter int unsigned WIDTH = 8`
Port	`a_i : in std_logic_vector(7 downto 0)`	`input logic [7:0] a_i`
Bit, vector	`std_logic`, `std_logic_vector`	`logic`, `logic [N-1:0]`
All zeros	`(others => '0')`	`'0`
Concatenation	`a & b`	`{a, b}`
Conditional	`y <= a when s = '1' else b;`	`assign y = s ? a : b;`
Combinational	`process(a, b)`	`always_comb`
Clocked	`process(rst_i, clk_i)` + `rising_edge`	`always_ff @(posedge clk_i or negedge rst_ni)`
Enumeration	`type t_state is (IDLE, RUN);`	`typedef enum logic {StIdle, StRun} state_e;`
Instance	`u0: entity work.adder port map(...)`	`adder u0 (...);`
Width of N	`natural(ceil(log2(real(N))))`	`$clog2(N)`

Modules, Ports, and Parameters

VHDL

```vhdl
library ieee;
use ieee.std_logic_1164.all;
use ieee.numeric_std.all;
entity adder is
  generic(
    WIDTH : positive := 8
  );
  port(
    a_i   : in  std_logic_vector(WIDTH - 1 downto 0);
    b_i   : in  std_logic_vector(WIDTH - 1 downto 0);
    sum_o : out std_logic_vector(WIDTH downto 0)
  );
end entity adder;
architecture rtl of adder is
begin
  sum_o <= std_logic_vector(
    resize(unsigned(a_i), WIDTH + 1)
    + unsigned(b_i));
end architecture rtl;
```

SystemVerilog

```systemverilog
`default_nettype none
module adder #(
  parameter int unsigned WIDTH = 8
) (
  input  logic [WIDTH-1:0] a_i,
  input  logic [WIDTH-1:0] b_i,
  output logic [WIDTH:0]   sum_o
);
  assign sum_o = {1'b0, a_i} + {1'b0, b_i};
endmodule
`default_nettype wire
```

- No type conversions
- `` `default_nettype none ``
  - a typo becomes an error
- Case-sensitive, one module per file

Data Types: logic

- `logic`: four values `0`, `1`, `x`, `z`, for **every RTL signal**
- **One driver** per variable (Verilator: `MULTIDRIVEN`)
- `reg` is **not a register**: storage follows from how a variable is assigned
- Vectors: `logic [7:0] data`, `data[7:4]`, `data[i*4 +: 4]`
- Unsigned by default, `logic signed [15:0]`: **do not mix** them
- Two-state `bit`, `int`: parameters, loop variables, testbenches only
- Icarus is four-state (x visible), Verilator two-state: **simulate with both**

Literals

VHDL	SystemVerilog	Meaning
`"10100101"`	`8'b1010_0101`	8-bit binary, _ for readability
`x"A5"`	`8'hA5`	8-bit hexadecimal
`std_logic_vector(to_unsigned(165, 8))`	`8'd165`	8-bit decimal
`(others => '0')`, `(others => '1')`	`'0`, `'1`	all bits zero or one, any width
`(others => 'X')`, `(others => 'Z')`	`'x`, `'z`	all bits unknown or high impedance
`"10X1"`	`4'b10x1`	a literal with an unknown bit
`165`	`165`	unsized: a 32-bit signed integer

Always size your literals.

Operators

VHDL	SystemVerilog	Meaning
`not a`	`~a`	bitwise NOT
`a and b`, `a or b`, `a xor b`	`a & b`, `a | b`, `a ^ b`	bitwise AND, OR, XOR
`and a` (VHDL-2008)	`&a`, `|a`, `^a`	reduction, 1-bit result
`not`, `and`, `or` on `boolean`	`!a`, `a && b`, `a || b`	logical, 1-bit result
`=`, `/=`	`==`, `!=`	equality, can be x with x operands
`<`, `<=`, `>`, `>=`	`<`, `<=`, `>`, `>=`	relational
none	`===`, `!==`	case equality, testbenches only
`shift_left()`, `shift_right()`	`<<`, `>>`, `>>>`	logical, arithmetic shift
`a & b`	`{a, b}`, `{4{a}}`	concatenation, replication
`a when s = '1' else b`	`s ? a : b`	conditional operator

Expression Widths

```
logic [7:0] a, b, sum8;
logic [8:0] sum9;
logic [3:0] nibble, count;
logic       hit;

assign sum8   = a + b;          // carry silently lost, no warning
assign sum9   = a + b;          // carry kept, the target is 9 bits wide
assign nibble = a;              // MSBs cut off (Verilator: WIDTHTRUNC)
assign hit    = (count == 20);  // always false (Verilator: WIDTHEXPAND)
```

- Operands extend to the widest operand **and the target**, then truncate: **no error**
- Verilator catches the truncation and the impossible compare, **not the lost carry**
- Make extensions explicit: `{1'b0, a}` or `9'(a)`

Concurrent Assignments

```
assign y_o   = a_i & b_i;           // VHDL: y_o <= a_i and b_i;
assign mux_o = sel_i ? a_i : b_i;   // VHDL: mux_o <= a_i when sel_i = '1' else b_i;
```

- `assign` is re-evaluated whenever an operand changes
- `with ... select` becomes a `case` in `always_comb`
- Output ports can be **read** inside the module: no internal copy needed

Exercise (Section 2.5.4): translate the VHDL 4:1 multiplexer `mux4` into SystemVerilog and lint it with `verilator --lint-only -Wall`.

Processes

- `always_comb`: combinational logic
  - complete sensitivity list inferred, like `process(all)`, the tools check for latches
- `always_ff @(posedge clk_i or negedge rst_ni)`: flip-flops
  - only the clock edge and the reset edge in the event list
- `always_latch`: an intended latch (rarely needed)
- Two assignments:
  - **blocking** (=): updates immediately, like a VHDL variable `:=`
  - **non-blocking** (<=): updates after all blocks have read their inputs, like a VHDL signal

Blocking vs. Non-Blocking

```
always_ff @(posedge clk_i) begin
  a_q <= b_q;
  b_q <= a_q;  // reads the old a_q: a real swap
end
```

- With = in this block: b_q reads the **new** a_q, both end up with the old b_q: **no swap**
- With = in **two blocks**: the result depends on the order in which the simulator runs the blocks, synthesis builds the swap: **simulation and hardware differ**

The Golden Rules

1. `always_ff`: only `<=`
2. `always_comb`: only `=`
3. One block per variable
4. No plain `always` or `initial` in RTL

Registers and Resets

```
// Asynchronous active-low reset (course default)
always_ff @(posedge clk_i or negedge rst_ni) begin
  if (!rst_ni) begin
    q <= '0;
  end else begin
    q <= d;
  end
end

// Synchronous active-high reset (template counter)
always_ff @(posedge clk_i) begin
  if (reset_i) begin
    q <= '0;
  end else begin
    q <= d;
  end
end
```

- Default: async active-low `rst_ni`
- Matches the cells: `sg13cmos5l_dfrbp_1` has an active-low `RESET_B`
- Reset test first, polarity matches the event list
- Assert asynchronously, **release synchronously** (Chapter 9)
- No reset: only `posedge clk_i` in the event list
- Clock enable: a register, **not a latch**

The Two-Process Style in SystemVerilog

```
// Register process: only clock and reset
always_ff @(posedge clk_i or negedge rst_ni) begin
  if (!rst_ni) begin
    state_q <= 1'b0;
  end else begin
    state_q <= state_d;
  end
end

// Combinational process: list inferred
always_comb begin
  // default assignment
  state_d = state_q;

  // state_d from state_q and inputs
  if (toggle_i) begin
    state_d = ~state_q;
  end
end
```

- `state` / `next_state` become `state_q` / `state_d` (**Q** and **D**)
- `always_ff`: only clock and reset edge (data signal: Yosys error)
- `always_comb`: sensitivity list **inferred**
- **Default first**, else Verilator `LATCH` and a Yosys error
- Yosys: `$_DFFE_PN0P_` and an inverter, the default became the **enable**

Example: Tick Generator

VHDL: `counter_logic`

```vhdl
counter_logic: process(count, en_i) is
begin
  next_count <= count;
  next_tick <= '0';
  if en_i = '1' then
    if count = C_LAST then
      next_count <= (others => '0');
      next_tick <= '1';
    else
      next_count <= count + 1;
    end if;
  end if;
end process counter_logic;
```

SystemVerilog: `always_comb`

```systemverilog
always_comb begin
  count_d = count_q;
  tick_d  = 1'b0;
  if (en_i) begin
    if (count_q == COUNT_LAST) begin
      count_d = '0;
      tick_d  = 1'b1;
    end else begin
      count_d = count_q + 1'b1;
    end
  end
end
```

- `tick_o` is a **strobe**: high for one cycle every `DIV` cycles while `en_i` is high

if, case, and Latches

```
// Latch: y_o holds its value while en_i is low
always_comb begin
  if (en_i) y_o = a_i;
end

// Fixed: default assignment first
always_comb begin
  y_o = '0;
  if (en_i) y_o = a_i;
end
```

- A `case` need **not** be complete: a missing value means a **latch**
- `unique case` + `default`
- `casez` for decoders, avoid `casex`
- `always @(*)`: Yosys warns and builds a latch
- `always_comb`: Yosys **stops with an error**

Exercise: Let the Linter Find the Bugs

```
logic [7:0] alu;
logic [7:0] result_q;
// Result register
always_ff @(posedge clk_i or negedge rst_ni) begin
  if (!rst_ni) begin
    result_q = 8'h00;
  end else begin
    result_q = alu;
  end
end
// ALU: 00 add, 01 subtract, 10 AND, 11 OR
always_comb begin
  case (op_i)
    2'b00: alu = a_i + b_i;
    2'b01: alu = a_i - b_i;
    2'b10: alu = a_i & b_i;
  endcase
end
assign result_o = result_q;
assign zero = (result_q == 8'h00);
```

A small ALU: `op_i` selects add, subtract, AND, or OR. `result_o` is registered, `zero_o` flags a zero result.

- Lint it with Verilator (`-Wall`)
- Explain **every** warning
- Fix it until Verilator is silent
- Synthesize both versions with Yosys

Exercise (Section 2.5.8): lint, fix, and synthesize the full file.

Finite State Machines

```
typedef enum logic [1:0] {StIdle, StRun, StDone} state_e;
state_e state_d, state_q;

always_comb begin
  state_d = state_q;  // defaults first
  busy_o  = 1'b0;
  done_o  = 1'b0;
  unique case (state_q)
    StIdle: if (start_i) state_d = StRun;
    StRun: begin
      busy_o = 1'b1;
      if (last_i) state_d = StDone;
    end
    StDone: begin
      done_o  = 1'b1;
      state_d = StIdle;
    end
    default: state_d = StIdle;  // unused 2'b11
  endcase
end
```

- VHDL: `type t_state is (...)`, the same `fsm_logic` structure
- `typedef enum` with base type, `_e`, `St` prefix
- Unused `2'b11`: `default`
- **Defaults first**: state and outputs
- Transitions with `if`/`else` (Icarus rejects `?:` between enums)
- Yosys re-encodes one-hot

Exercise (Section 2.5.9): the controller as a Mealy FSM.

Hierarchy and Generate

VHDL

```
adder_0: entity work.adder
  generic map(
    WIDTH => 16
  )
  port map(
    a_i => x,
    b_i => y,
    sum_o => s
  );
```

- Connect **by name**, every port
- No `.*` or `.name`
- Instance name: `gen_lane[2].adder_i`

SystemVerilog

```
adder #(
  .WIDTH(16)
) adder_0 (
  .a_i  (x),
  .b_i  (y),
  .sum_o(s)
);
```

```
for (genvar i = 0; i < 4; i++) begin : gen_lane
  adder #(
    .WIDTH(8)
  ) adder_i (
    .a_i  (lane_a[i]),
    .b_i  (lane_b[i]),
    .sum_o(lane_sum[i])
  );
end
```

Packages, Typedefs, and Functions

```
package alu_pkg;
  localparam int unsigned DATA_WIDTH = 8;

  typedef enum logic [1:0] {
    OpAdd, OpSub, OpAnd, OpOr
  } alu_op_e;

  typedef struct packed {
    alu_op_e                  op;
    logic [DATA_WIDTH-1:0] operand;
  } instr_t;

  function automatic logic is_zero(
      input logic [DATA_WIDTH-1:0] data);
    is_zero = (data == '0);
  endfunction
endpackage
```

- Use: `alu_pkg::OpAdd`, `alu_pkg::is_zero(result_o)`
- `struct packed` replaces `record`
- `function automatic`: combinational logic, one copy per call
- Compile the package **first**

Yosys 0.70 default front end (LibreLane): no `import alu_pkg::*`, no `return`, no unpacked array ports.

Use `pkg::NAME`, or set `USE_SLANG: True`.

What Is Synthesizable?

Synthesizable (RTL)	Simulation and testbenches only
`module`, `parameter`, `localparam`, `assign`	`initial` (ASICs have no initial values: use a reset)
`always_comb`, `always_ff`, `always_latch`	delays `#10`, event waits `@(...)` and `wait`
`if`, `case`, `for` with constant bounds	`$display`, `$finish`, `$fatal`, `$dumpvars`
`function automatic`, generate	`===`, `!==`, `real`, `string`
`typedef enum`, `struct packed`, packages	classes, `randomize()`, `fork` and `join`
`+`, `-`, `*`, shifts, comparisons	loops with data-dependent iterations

`/` and `%`: large and slow, unless the divisor is a constant power of two

Verilog vs. SystemVerilog

Aspect	Plain Verilog	SystemVerilog
Signals	`wire` for `assign`, `reg` for `always`	`logic` for both
Ports	`module m (a, y); input a; ...`	`module m (input logic a_i, ...);`
Combinational	`always @(*)`: a latch is a Yosys warning	`always_comb`: a latch is a Yosys **error**
Flip-flops	`always @(posedge clk or negedge rst_n)`	`always_ff @(posedge clk_i or negedge rst_ni)`
Constants	`` `define ``, a `localparam` per state	`package`, `typedef enum` for the states
All zeros	`{W{1'b0}}`	`'0`
Loops	`integer i;` … `i = i + 1`	`for (int i = 0; i < N; i++)`
Case checks	`// synopsys full_case` comment	`unique case` + `default`
Tools	`.v`, `read_verilog`, `iverilog`	`.sv`, `read_verilog -sv`, `iverilog -g2012`

Read Verilog (netlists, cell models, generated code, older IP), **write** SystemVerilog

Exercise (Section 2.5.13): modernize a Verilog-2001 priority encoder.

Testbenches



A SystemVerilog Testbench

```
always #(CLK_PERIOD_NS / 2) clk = ~clk;

initial begin
  $dumpfile("tick_gen_tb.vcd");
  $dumpvars(0, tick_gen_tb);
  repeat (2) @(negedge clk);
  rst_n = 1'b1;
  // ... 3 * DIV cycles with en = 0: no tick ...
  en = 1'b1;
  for (int cycle = 1; cycle <= 4 * DIV; cycle++) begin
    @(negedge clk);
    if (tick !== (cycle % DIV == 0)) begin
      $fatal(1, "FAIL: cycle %0d, tick = %b", cycle, tick);
    end
  end
  $display("PASS: tick_gen_tb");
  $finish;
end
```

- Module **without ports**, DUT with a small `DIV`
- Drive and check on the **falling edge**
- `!==` also fails on x
- Fail with `$fatal(1, ...)`
  - `$error` exits with 0 in Icarus
- `$dumpvars`: VCD waveform

A cocotb Testbench

```python
@cocotb.test()
async def test_tick_every_div_cycles(dut):
    """tick_o is high for one cycle every DIV cycles."""
    await start_up(dut)  # 50 MHz clock, reset for two cycles
    dut.en_i.value = 1
    for cycle in range(1, 4 * DIV + 1):
        await FallingEdge(dut.clk_i)
        expected = 1 if cycle % DIV == 0 else 0
        assert dut.tick_o.value == expected, \
            f"cycle {cycle}: expected {expected}, got {dut.tick_o.value}"

# end of the runner: a failing test must set the exit status
num_tests, num_failed = get_results(results_xml)
if num_failed:
    raise SystemExit(f"FAIL: {num_failed} of {num_tests} tests failed")
```

- Tests are coroutines: `await FallingEdge(...)`, `ClockCycles(...)`, `Timer(...)`
- A write takes effect at the **end of the time step**

SystemVerilog or cocotb?

Aspect	SystemVerilog testbench	cocotb testbench
Language	SystemVerilog, the full language	Python 3
Reference models	written in SystemVerilog	any Python code and library (NumPy)
Simulators	every SystemVerilog simulator	Icarus, Verilator, GHDL, commercial
DUT languages	SystemVerilog, VHDL only with a mixed-language simulator	VHDL and SystemVerilog alike
Speed	fast, all in the simulator	slower, every access goes through Python
Template folder	`testbenches/verilog/`	`testbenches/cocotb/`

Both: SystemVerilog for a **quick self-check**, cocotb for **models and random stimuli**

Lint, Simulate, and Synthesize



Linting with Verilator

```
verilator --lint-only -Wall rtl/tick_gen.sv
```

Warning	Meaning	Typical fix
`IMPLICIT`	undeclared name, an implicit 1-bit wire	fix the typo, `` `default_nettype none ``
`UNDRIVEN`, `UNUSEDSIGNAL`	never assigned or never read	connect or remove it
`WIDTHTRUNC`, `WIDTHEXPAND`	widths do not match	sized literals, `{1'b0, x}`, `N'(x)`
`LATCH`	a combinational block misses a path	defaults first
`CASEINCOMPLETE`	`case` misses values	add the missing items or a `default`
`BLKSEQ`	= in a sequential block	<= in `always_ff`
`MULTIDRIVEN`	driven from two blocks	exactly one block
`DECLFILENAME`	file name differs from module	one module per file

Simulation and Waveforms

```
# SystemVerilog testbench with Icarus Verilog
iverilog -g2012 -o testbenches/verilog/tick_gen_tb.vvp \
    rtl/tick_gen.sv testbenches/verilog/tick_gen_tb.sv
(cd testbenches/verilog && vvp tick_gen_tb.vvp)

# cocotb testbench with Icarus Verilog, then with Verilator
(cd testbenches/cocotb && python3 tick_gen_tb.py)
(cd testbenches/cocotb && SIM=verilator python3 tick_gen_tb.py)

# Waveforms
gtkwave testbenches/verilog/tick_gen_tb.vcd
surfer testbenches/cocotb/sim_build/tick_gen.fst
```

A Quick Synthesis Check with Yosys

```
yosys -p "read_verilog -sv rtl/tick_gen.sv; synth -top tick_gen; stat"
```

- Flip-flops (`$_DFF_*`): as many as the **registers you intended**
- Latches (`$_DLATCH_*`): **none**
- Tick generator (`DIV = 10`): 17 cells, 4 × `$_DFFE_PN0P_` and 1 × `$_DFF_PN0_`
  - `PN0P`: positive clock, active-low reset to 0, active-high enable

Live Demo: Tick Generator

```
mkdir -p /foss/designs/sv_exercises/tick_gen/{rtl,testbenches/verilog,testbenches/cocotb}
cd /foss/designs/sv_exercises/tick_gen
```

- Save `rtl/tick_gen.sv` and both testbenches (Sections 2.5.7, 2.6)
- **Lint**, **simulate** (SystemVerilog and cocotb), look at the **waves**, **synthesize**
- Build in a bug (`COUNT_LAST - 1`): do both testbenches catch it?

Exercise (Section 2.7.6): the tick generator, then a PWM generator with a SystemVerilog and a cocotb testbench, and a sequence detector FSM.

Hands-On: The Template Counter

```
cd /foss/designs
git clone https://github.com/iic-jku/ihp-sg13cmos5l-ams-chip-template.git
cd ihp-sg13cmos5l-ams-chip-template/macros/counter
```

- `macros/counter`: 8-bit counter, synchronous reset, enable, wrap at `CTR_MAX`
- Lint and simulate it **by hand**: what does each testbench check?
- Rewrite it in the **two-process style**, build in a bug, check the exit status
- What is behind the **make targets**?
  - `make help`, `make -n sim-rtl-verilog`

AMS chip template

Exercise (Section 2.8): three exercises with the template counter.

Guidelines and Summary



HDL Coding Guidelines

"A consistent naming and schematic drawing style, as well as VHDL/Verilog coding scheme, is a huge help in avoiding errors and increasing productivity." (Circuit Designer's Etiquette)

- **Lint** with `-Wall`, fix every warning
- `_i`, `_o`, `_io`, active low: `n` before it (`rst_ni`)
- `logic` everywhere, explicit ports
- `localparam`, parameters in UPPERCASE
- **Reset** every control register
- **One module per file**, same name
- `` `default_nettype none `` ... `wire`
- `always_comb`, never `always @(*)`
- Explicit **widths** and conversions
- `=` in `always_comb`, `<=` in `always_ff`

From lowRISC: `_d`/`_q`, `_e`/`_t` types, `St` states, `unique case` + `default`, named connections

Pitfalls for VHDL Designers

Pitfall	What happens	Fix
expecting `reg` to be a register	storage follows from the block	use `logic`
`=` in `always_ff`	order-dependent, simulation differs from synthesis	`<=` in `always_ff`
missing `else` or `default`	a latch	defaults first
`Clk` and `clk`	two different signals	lowercase snake_case
a typo in a name	an implicit 1-bit wire	`` `default_nettype none ``
mismatched widths	silent truncation, a lost carry	sized literals, lint
`==` with `x` in a testbench	can be `x`, the check passes	`!==`, `$isunknown()`
`initial` for reset values	not synthesizable for an ASIC	reset in `always_ff`
driving on the active edge	races, simulators differ	drive and check on the falling edge
`$error` in Icarus Verilog	exit status 0	`$fatal(1, ...)`

Summary

- An HDL describes **concurrent hardware**: think hardware first
- Register + next-state logic: **one clock period** to compute
- Alternative HDLs end as **(System)Verilog**: **read** Verilog, **write** SystemVerilog
- Your **two-process style** maps directly: `always_ff` copies `_d` into `_q`, `always_comb` holds the defaults and all decisions
- `logic`, `=` in `always_comb`, `<=` in `always_ff`, sized literals, **lint everything**
- **Self-checking** testbenches that set the **exit status**: `$fatal`, `get_results()`

Exercises

1. Loops in Software and in Hardware (2.2.1)
2. A 4:1 Multiplexer in SystemVerilog (2.5.4)
3. Let the Linter Find the Bugs (2.5.8)
4. The Controller as a Mealy FSM (2.5.9)
5. Modernize a Verilog-2001 Module (2.5.13)
6. Lint, Simulate, and Synthesize the Tick Generator (2.7.6)
7. PWM Generator with a SystemVerilog Testbench (2.7.6)
8. PWM Generator with a cocotb Testbench (2.7.6)
9. Sequence Detector FSM (2.7.6)
10. Lint and Simulate the Counter by Hand (2.8)
11. The Counter in the Two-Process Style (2.8)
12. What Is Behind the Make Targets? (2.8)



Chapter 2



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Backup: Alternative HDLs at a Glance

Family	Host language	Languages (selected origins)
Embedded DSL	Scala	Chisel (UC Berkeley), SpinalHDL, DFiant
Embedded DSL	Python	Amaranth, Migen (M-Labs), PyMTL3 (Cornell), PyRTL, HWT, Magma and Kratos (inactive)
Embedded DSL	Haskell, OCaml	Clash (University of Twente), Hardcaml (Jane Street)
Embedded DSL	Dart, Rust	ROHD (Intel), RHDL
Standalone HDL	own syntax	Bluespec, Veryl, Spade, Silice
HLS, compiler IR	DSLX, C/C++, IR	XLS (Google), Calyx (Cornell), Vitis HLS, Catapult, Bambu